

SHRI G.S. INSTITUTE OF TECHNOLOGY AND SCIENCE, INDORE - 3

Name : HARSH RAJ SHARMA

CARPENTRY - SHOP

Roll No. : AB-16030

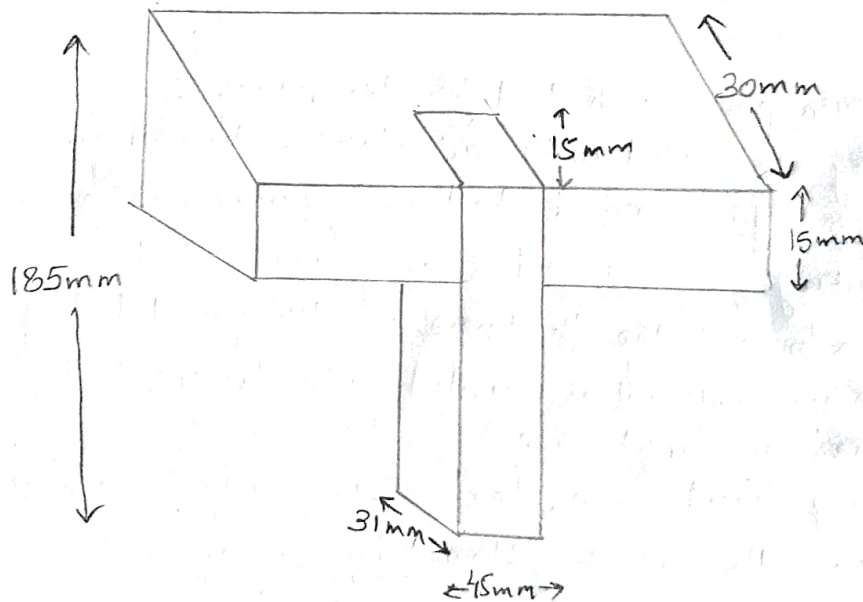
TITLE : To prepare a wooden joint as per drawing.

OBJECTIVES : To aware with the various wood, wood working operation and respective tools used for the operation.

: Awareness of various type of wood.

: Various types of wood joints.

DRAWING:



Raw material size :

Length = 300 mm

width = 50 mm

Thickness = 35 mm

All dia in m.m.

Raw material :-

Teak wood

Hand Tool/Cutting Tools	Particular	Specification
Hand Tools	Hand saw, jack plane Mallet, jermer chisel	High grade steel mild carbon steel Cross cut saw length - 22" long
Cutting Tools	Cross cut hand, saw jermer, chisel, jack plane	High grade steel, blade & steel.
Measuring/ Marking Tools used	Foot rule, Try square, Marking guage, Mortise guage	Cast iron, Mark, Parallel linear

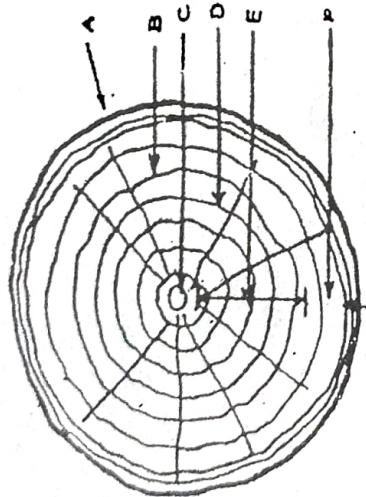
Sequence of operations :

- 1.) Given piece is checked for dimensions.
- 2.) They are placed with Jack plane & checked for straightness.
- 3.) Two surfaces are checked for squareness with try square.
- 4.) Marking guage is set & lines are marked at 30 mm and 45 mm to mark the thickness & width of model respectively.
- 5.) Excess material is chiseled with trimmer & then planed to correct size.
- 6.) Using hand-saw, the parts to be removed are cut in both pieces followed by chiseling.

Precautions:

- 1.) Tools are to be placed at their proper places.
- 2.) Hands should not be placed in front of sharp edged tools.
- 3.) Use only sharp tools.
- 4.) Loose cloths are to be avoided.
- 5.) Care should be taken when thumb is used as a guide in cross cutting & ripping.
- 6.) Handle while chiseling, planing with care.

A cross section of a log of wood is shown below. Name the parts which are marked A, B, C etc. in the spaces provided.
 (First colour is filled as an example.)

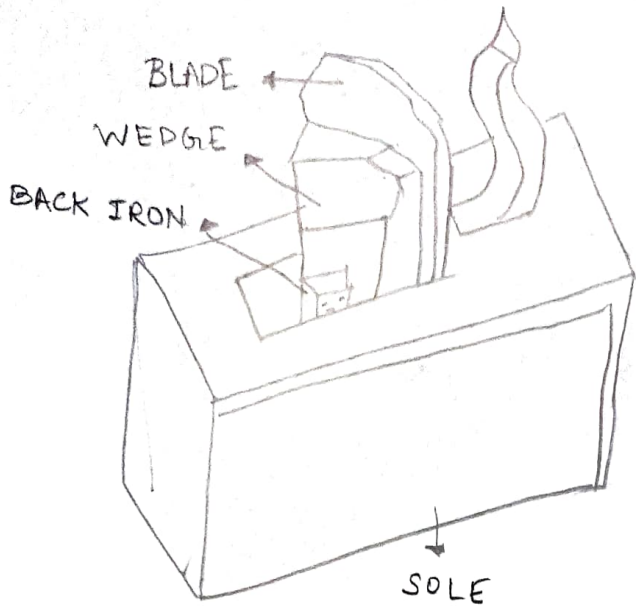


A	bARK
B	Annual Ring
C	Pitch
D	Rays
E	Heart wood
F	Sap wood

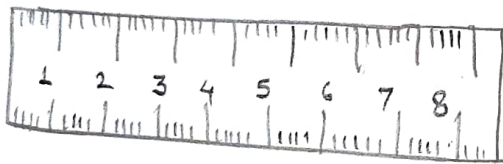
Give the names of the following sections of man-made boards in the spaces provided.

Section	Name	Application
	Black wood	furniture making
	Ply wood	furniture making, Door & cabin
	Balustr Board	furniture making
	Black Board	furniture making

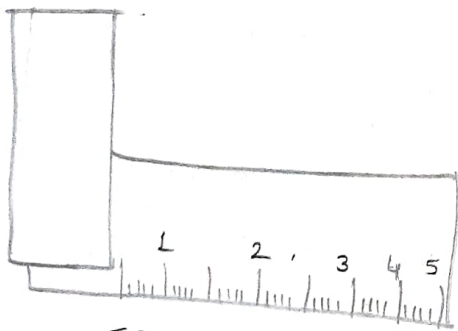
TOOLS USED IN CARPENTRY SHOP



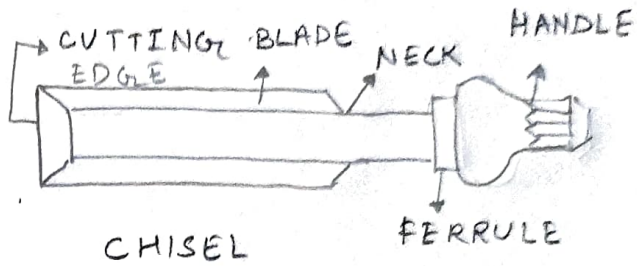
JACK PLANE



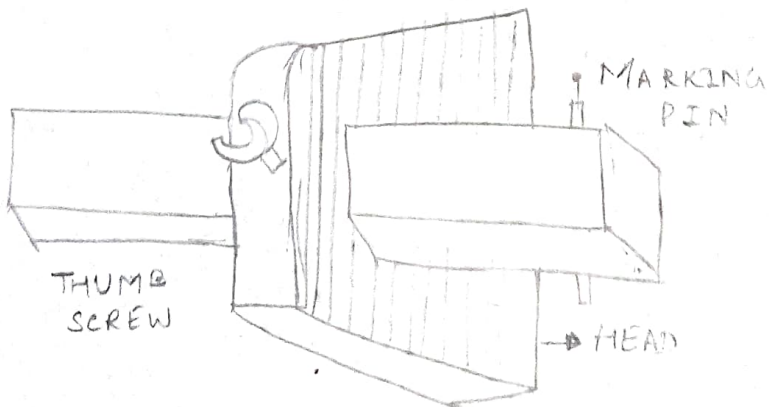
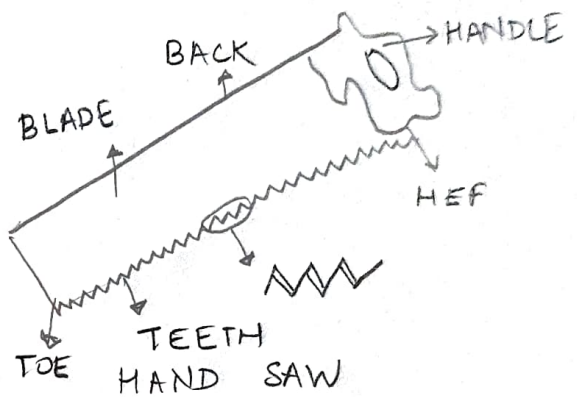
FOOT RULE



TRY SQUARE

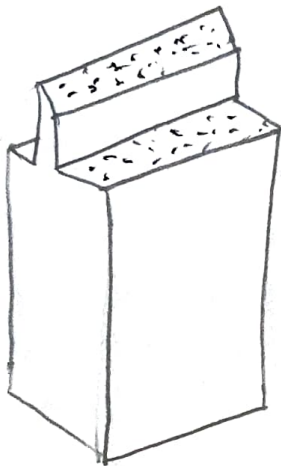
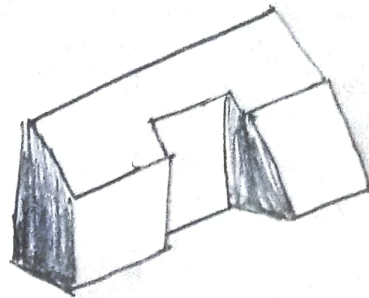
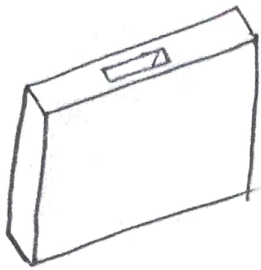


CHISEL

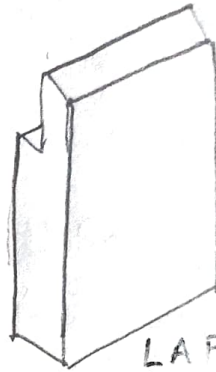


MARKING GAUGE

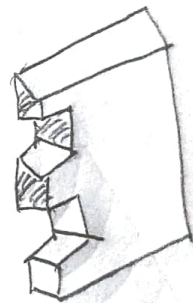
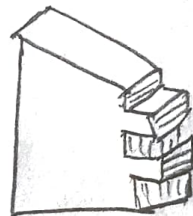
TYPES OF JOINTS



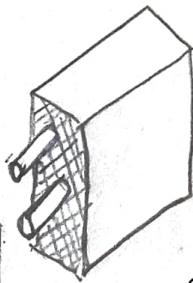
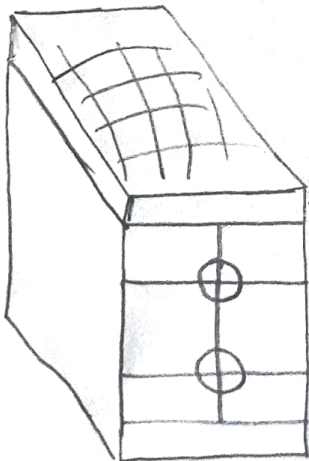
MORTISE/TENON



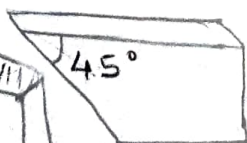
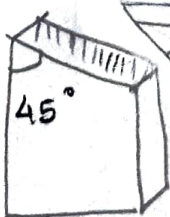
LAP JOINT



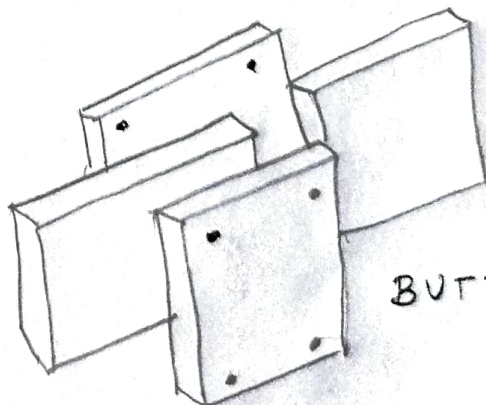
DOVE TAIL JOINT



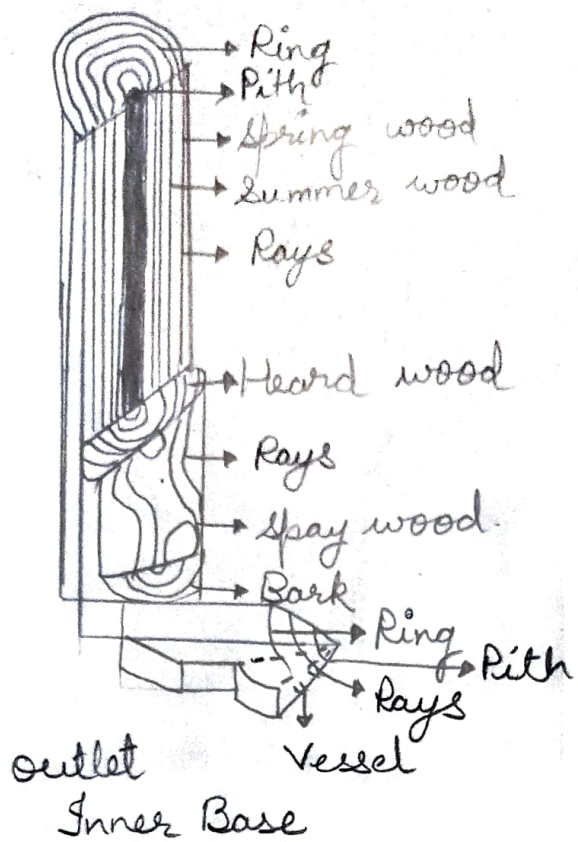
DOWEL JOINT



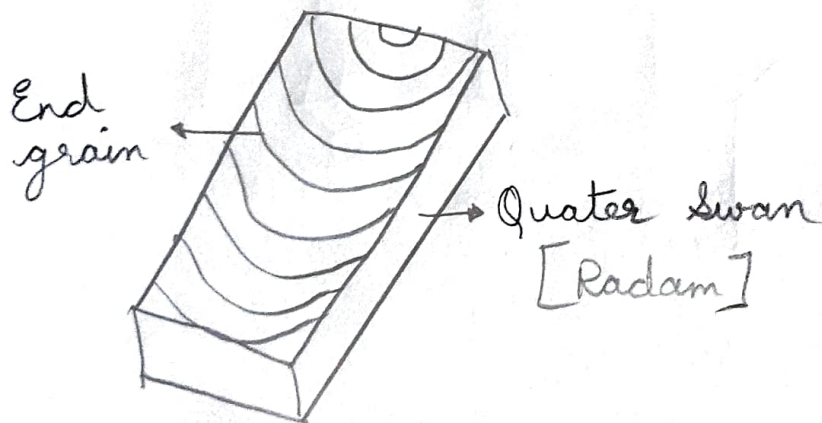
MITRE JOINT



BUTT JOINT



Wood Technology



QUESTIONARIES :

1. Name various types of woods and their uses.
2. Name various type of wood joint and write its application.
3. What is different between Block Board and plywood.
4. Write down the various cutting tools used in carpentry along with tool material Geomentry.
5. What kind of operations can be performed on universal wood working machine.
6. List various machines used in carpentry shop and their use.
7. What is the commercial unit of wood used in market.
8. Complete the sheet as enclosed herewith cross section of log of wood.
9. Give atleast two different type of Timber used for following purpose in this region.
 - (a) Ship Building
 - (b) Railway sleeper
 - (c) Furniture making
 - (d) Making door and windows
 - (e) Packing case
10. State whether the following statement are TRUE or FALSE by marking (T) or (F)
 - (a) Soft woods are light in weight
 - (b) Hard woods are darker in colour.
 - (c) Soft woods can successeffuly resists the load which are applied across their grains.
 - (d) Across the grain means axial to the cross section of the trunk.
 - (e) Hard woods are stranger when loads are applied along the grains.

1.) Mainly, there are two varieties of woods' -

(i) Hard wood' -

a.) Mahogany' - Reddish-brown - & fine grained

Best used for cabinetry and board construction.

b.) Rosewood' - Dark reddish-brown and very hard.

It is closely grained & it suits to work on used for musical instruments -

c.) Walnut' - Fine textured & strong.

Resists shrinking and warping. Used for cabinetry, furniture, gunstocks & wall paneling.

d.) Bak' - Strong yet bendable. Used for boat framing, flooring, furniture & tool handles.

(ii) Softwood' -

a.) Pines' - Has uniform texture, easy to work on.

Used for constructing houses, packing boxes & moulding.

b.) Hemlock' - light weight, uniformly textured, used for doors, frames, plywood & construction timber.

c.) Fir' - Low resistance to decay. Best used for doors, furnitures, frames, trusses.

d.) Spruce' - It is very hard & strong. It decays easily, so that wood protection is need to prevent dry or wet rot shrinks mandatory, used for spars, masts in ships, crates, boxes, ladders & general millwork.

② Various types of wood joints are -

- (i) Mortise and tenon joints - Commonly used by wood workers, consists of a rectangular peg fitting into a rectangular hole (mortise). Tools used in particular are mortise, chisel & mortise gauge.
- (ii) Bridle joint - Reverse of mortise & tenon joint. Also call open mortise & tenon. There are two grooves and a slot.
- (iii) Lap joint - Called halving joint, secures the corners & intersection of framing, used in many types of framing.
- (iv) Dowetail joint - Strongest of all corner joints. Takes strain in one direction but has several variations. Used in making large pieces of furnitures, boxes & cases.
- (v) Dowel Joint - The Dowelled joint is often used as forming joint in place of mortise & tenon joint. Used for circular work, but joining edges or furnishing movable fittings.

→ Types of classified joints: -

- Lap joints: - Cross lap, corner, dovetail lap
- Bridle joint: - T-Bridge, corner bridge, butt joint
- Dovetail: - Common, lapped, cupped secret, secret
- Mortise/Tenon: - Simple, blind, double, haunched

Black Board

- ③ (i) Black board has a timber frame with solid wood bottom and is light in weight.
- (ii) Black board is a solid, timber filled sandwiched b/w two layers of veneers.
- (iii) May have minor gaps between them.
- (iv) Usually have a core made of softwood blocks.
- (v) Inferior quality.

Plywood

- (i) Solid in nature & made of timber layers.
- (ii) Plywood is made up of several layers of veneers (thin sheet).
- (iii) Uniform throughout its length & breadth.
- (iv) Made using hardwood knees, heavier than softwood.
- (v) Superior quality.

④ Cutting tools include:-

- Saw - Chisel - Gauge - Router - Lathe

⑤ Universal wood working machine, i.e. the wood working lathe is one of the most important machine in carpentry shop.

Primarily used for turning jobs in making cylindrical parts, but by manipulating tools, tapers, radii & other shapes can also be turned.

Consists of cast iron bed, head stock, tail stock, tool rest, live and dead centers & a speed control device.

Following operations can also be performed on this machine:-

- | | | |
|---------------|---------------------|-------------------|
| (i) Drilling | (ii) Blade grinding | (iii) Multisizing |
| (iv) Grooving | (v) Pattern cutting | (vi) Moulding |

(6) The machines used in carpentry shop & their uses are as follows:-

(i) Wood working lathe:- Primarily used for turning jobs in making cylindrical parts, but by manipulating tools, tapers, square & other shapes can also be turned. Used in drilling, mortising, turning, taper turning, grooving, blade grinding, pattern cutting & moulding.

(ii) Band saw:- Designed to cut wood by means of an endless metal saw band that travels axis the rims of two more rotating wheels.

Other parts of a band saw are frame, table, saw guides, etc. Mostly used in making curved or irregular arts in wood.

(iii) Circular saw:- Second most important machine in carpentry shop. Used for ripping, cross cutting, mitring, benching, rabbeting & grooving.

Different types of circular saw are universal saw, variety saw, hand saw, etc. Usually hand variety saw, are provided with a tilt the table to an angle of 45° to enable the machine to cut at different angles required during mitring, levelling etc.

(iv) Jointer:- Used for planing straight edges and surfaces eg board, performs the work of a hand plane & capable of producing a true surface with sufficient accuracy and speed can also be used for angular and level cuts.

(v) Wood planes:- Designed for planing large & heavy stock at faster rate. The table of the plane is much wider & longer than that of a jointer for accommodating large & heavy stock.

(vi.) Mortiser - A mortiser is a narrow slot cut in the direction of depth from for the purpose of making a mortise & tenon joint in wooden piece.

(vii.) Sanding machines - Sanding is the operation of finishing wooden item, after they have been machined.

Essentially, a sanding machine performs a sand papering jobs to provide a uniformly sanded surface.

(viii.) Grinder - A small tool, grinder is used for sharpening & shaping various tool in a wood working shop. Grinder has two grinding wheels fastened to the two ends of a revolving spindle which is driven by a small electric motor. ~~It~~ ^{one} is used for coarse grinding & the other for fine grinding.

(7) There are many units used for measurement of wood in market. Log scales are used for pulpwood. Used for paper & fuel is the 'cord'. One cubic foot of wood (volume) is taken standard volume.

(9) Ship Building:- Teak, ~~widely~~ available in Central Mahogany India]

(ii) Railway sleepers? - Oak & Mahogany, Deodar.

(iii) Furniture making? - Fir, Mango & Walnut.

(iv) Making door & windows? - Fir, Teak & Sal.

(v) Packing case? - Pine, Deodar.

Differences between Hardwood & Softwood -

<u>Softwood</u>	<u>Hardwood</u>
(i) Belongs to conifers which have long narrow leaves.	(i) Belong to broad leaved trees.
(ii) Contains turpentine oil, resin, matter & lignin	(ii) Contains lignin, cellulose, hemicellulose.
(iii) Softwoods are light in weight and light coloured	(iii) Darker in colour & are comparatively heavy.
(iv) More distinct annual rings but no visible medullary rays	(iv) Annual rings are more compact, thin & less distinct, medullary rays are visible in most.
(v) Fibres are generally less dense, but straight	(v) Fibres are fine grained, compact & properly banded
(vi) Capable of resisting direct axial stress, across their fibres.	(vi) Can resist axial stress as well as transverse stress, shock & vibration satisfactorily.

- 10
- a.) True
 - b.) True
 - c.) False
 - d.) False
 - e.) True